CA-GAR 是一种通过上下文感知来优化 LLM 生成内容以提升检索性能的新方法。它通过分布对齐策略和加权解码目标，成功克服了生成式检索模型中常见的分布失真问题

1. 文章的 Pipeline（流程）

a. 输入与初始化：接收原始查询（Query）和指令（Instruct）

b. 语料库信息提取：使用基于词典的检索方法（如 BM25）从目标语料库中提取与当前生成状态相关的候选文档

c. 上下文感知解码（核心步骤）：在 LLM 的每一步自回归解码（Autoregressive Decoding）中，通过 分布对齐策略（Distribution Alignment Strategy） 优化 Token 选择。具体来说，它将 LLM 的对数概率与基于 BM25 的文档相似度得分进行加权合并（由参数 β 控制），从而引导模型选择更符合语料库特征的词汇

d. 查询增强：将生成的伪文档（Pseudo-document）与原始查询拼接，形成新的增强查询

e. 最终检索：使用增强后的查询在语料库上执行稀疏（BM25）或稠密检索，获得最终的相关文档列表

2. contribution
CA-GAR 旨在解决传统生成增强检索（GAR）中的 内容偏差（Content Bias） 问题

a. 通过在生成阶段实时引入语料库上下文，强制 LLM 生成的内容在语义和词汇分布上与目标文档集高度对齐

CA-GAR: Context-Aware Alignment of LLM Generation for Document Retrieval

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Abstract

Information retrieval has evolved from traditional sparse and dense retrieval methods to approaches driven by large language models (LLMs). Recent techniques, such as Generation-Augmented Retrieval (GAR) and Generative Document Retrieval (GDR), leverage LLMs to enhance retrieval but face key challenges: GAR's generated content may not always align with the target document corpus, while GDR limits the generative capacity of LLMs by constraining outputs to predefined document identifiers. To address these issues, we propose **C**ontext-**A**ware **G**eneration-**A**ugmented **R**etrieval (**CA-GAR**), which enhances LLMs by integrating corpus information into their generation process. CA-GAR optimizes token selection by incorporating relevant document information and leverages a Distribution Alignment Strategy to extract corpus information using a lexicon-based approach. Experimental evaluations on seven tasks from the BEIR benchmark and four non-English languages from Mr.TyDi demonstrate that CA-GAR outperforms existing methods.

1 Introduction

Information retrieval (IR) has become a critical component of natural language processing (NLP). The field has evolved from traditional sparse retrieval methods based on lexicon, such as TF-IDF and BM25 (Robertson and Zaragoza, 2009), to dense retrieval (Lee et al., 2019; Karpukhin et al., 2020) approaches powered by deep learning and pre-trained models like BERT (Devlin et al., 2019).

The advent of Large Language Models (LLMs) has significantly enhanced information retrieval by leveraging their advanced natural language understanding and generalization capabilities (Brown et al., 2020; Gao et al., 2023b; Zhu et al., 2023). Current research in this domain primarily revolves around two paradigms: (1) Generation-Augmented Retrieval (GAR), which enhances retrieval performance by utilizing LLMs' generative abilities to refine queries (Mao et al., 2021; Gao et al., 2023a; Wang et al., 2023; Jagerman et al., 2023); and (2) Generative Document Retrieval (GDR), which employs LLMs to generate document identifiers directly through constrained decoding, thereby encoding corpus-specific information into the retrieval process (Cao et al., 2021; Tay et al., 2022). However, both approaches present challenges. In GAR, a common strategy involves generating auxiliary text (e.g., query rewriting (Wang et al., 2023; Jagerman et al., 2023; Shen et al., 2024) or hypothetical document embeddings such as HyDE (Gao et al., 2023a)) to assist retrieval. However, the generated content may not always align well with the characteristics of the target document corpus, leading to suboptimal retrieval performance. As illustrated in Figure 1, while the language model can generate both *Passage 1* and *Passage 2* as plausible outputs, *Passage 2* is better suited for retrieval within the given document corpus. Although GDR ensures that the generated content exists within the document corpus through constrained decoding, effectively avoiding the retrieval mismatches observed in GAR, this approach inherently limits the

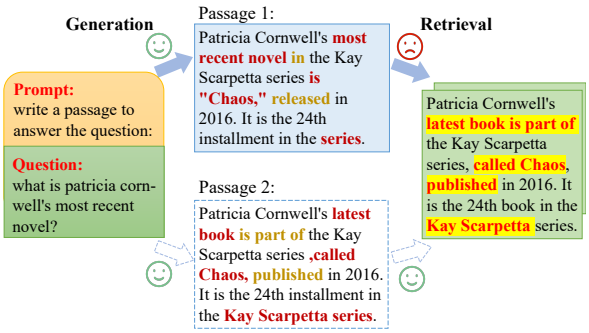

Figure 1: A figure illustrating how GAR's generated text may mismatch with the target document corpus.

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capacity of language models. (Li et al., 2024b).

Based on the observations above, our research question is: **How can we ensure that content generated by LLMs is well-suited for retrieval within a target document corpus while fully leveraging the generative capabilities of LLMs?** In retrieval tasks, the number of documents in the corpus is typically vast, far exceeding the context window that LLMs can process. Consequently, directly providing the entire document corpus as input to the model is infeasible. This presents a key challenge: **How can we effectively integrate corpus information into LLMs to enhance their retrieval performance?**

To address this challenge, we introduce a novel approach called **C**ontext-**A**ware **G**eneration-**A**ugmented **R**etrieval (**CA-GAR**), which incorporates corpus information into the generation process of LLMs. Specifically, at the core of our approach is the optimization of the model's autoregressive generation process by leveraging relevant document information from the corpus to influence token selection. To achieve this, we propose a **Distribution Alignment Strategy**, which utilizes a lexicon-based method to extract corpus information. This strategy approximates the optimization of the model's autoregressive generation process, ensuring that the generated content is better aligned with the target document corpus.

In summary, our contributions are as follows:

- We introduce a new approach called CA-GAR, which effectively combines the generative capabilities of LLMs with contextual information from the target document corpus.
- We propose a Distribution Alignment Strategy that utilizes a lexicon-based method to extract information from the corpus, optimizing autoregressive generation for improved alignment with the target document corpus.
- Our method outperforms existing approaches in retrieval tasks, as shown by experiments on seven BEIR benchmark tasks and four non-English languages from Mr. TyDi.

2 Related Work

Docuemnt Retrieval Information retrieval encompasses multiple domains and tasks (Liu et al., 2021; Zhuang et al., 2022; Sun et al., 2024). Our work primarily focuses on document retrieval. Early approaches were dominated by traditional lexicon-based methods, such as TF-IDF and BM25. These methods were later extended by sparse retrieval techniques that integrate neural networks with BM25, such as DeepCT (Dai and Callan, 2020) and docT5query (Nogueira et al., 2019). Recently, with the rise of pre-trained language models, dense retrieval (Lee et al., 2019; Karpukhin et al., 2020) has emerged as a promising alternative. By capturing semantic representations of text, dense retrieval effectively addresses the semantic mismatch problem inherent in sparse retrieval methods. Researchers have made significant advancements in this area through various techniques, including negative sample mining (Xiong et al., 2021; Qu et al., 2021), knowledge distillation (Qu et al., 2021; Lin et al., 2021; Ren et al., 2021; Liao et al., 2024), loss function optimization (Liao et al., 2024), and multi-vector representations (Zhang et al., 2022; Kang et al., 2025). However, dense retrieval still faces a critical challenge: achieving optimal performance often requires large-scale supervised training on domain-specific datasets, which limits its ability to generalize performance advantages across datasets from different domains (Thakur et al., 2021).

Large Language Models Large Language Models (LLMs) have demonstrated remarkable generalization capabilities, bringing significant transformations to the field of natural language processing (Brown et al., 2020; Ouyang et al., 2022; Hoffmann et al., 2022; Yuan et al., 2024). In the area of information retrieval, researchers have leveraged the instruction-following and in-context learning abilities of LLMs (Min et al., 2022; Sanh et al., 2022; Wei et al., 2022) to effectively address retrieval tasks by simply providing task definitions and a few retrieval examples (Li et al., 2024a). Building on this foundation, further studies have explored the use of LLMs for query enhancement, including query rewriting (Shen et al., 2024) and the expansion of relevant document information (Gao et al., 2023a). Notably, these approaches do not require additional training. Moreover, researchers have proposed parameter-efficient fine-tuning techniques to adapt portions of an LLM's parameters, enabling the model to function directly as an embedding model for retrieval tasks (Wang et al., 2024; BehnamGhader et al., 2024; Lee et al., 2024). This approach significantly reduces the computational and training resources required while preserving the LLMs' powerful natural language understanding capabilities.

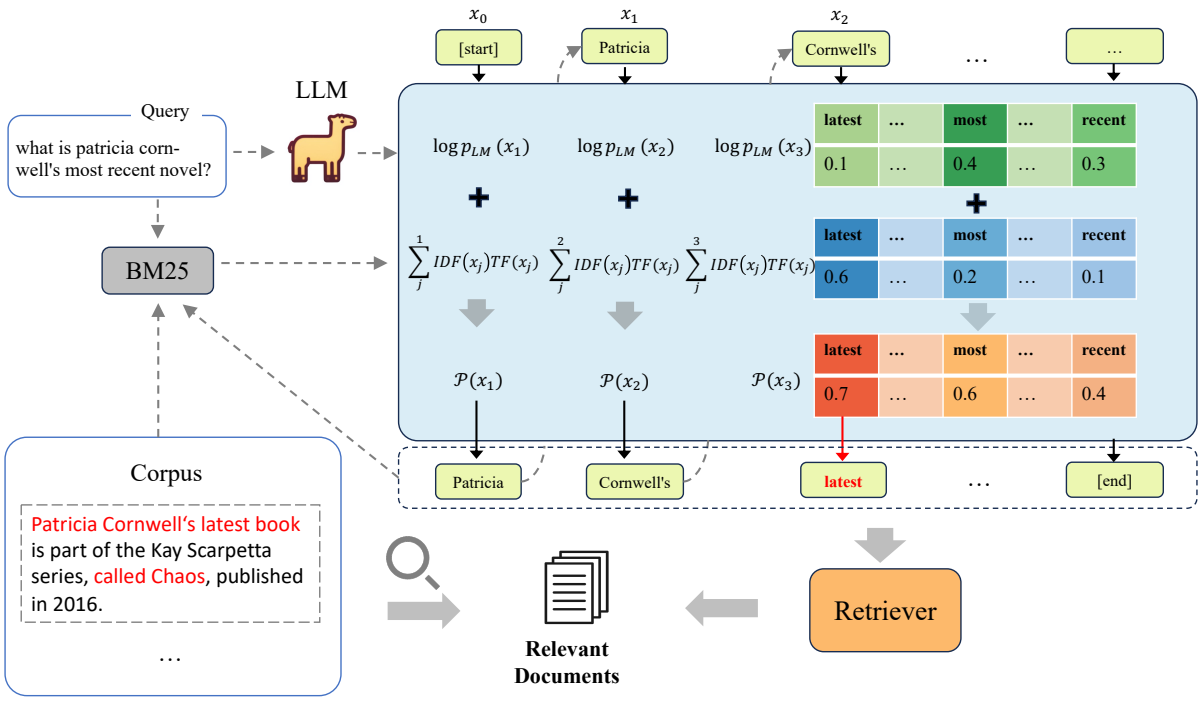

Figure 2: A figure illustrating our CA-GAR method.

Generative Retrieval Generative retrieval leverages the generative capabilities of models to directly facilitate document retrieval. Early approaches focused on enabling generative models to produce document identifiers through constrained decoding, which were then mapped to relevant documents (Cao et al., 2021; Tay et al., 2022). Subsequent research expanded on this paradigm by exploring techniques such as query augmentation and the design of more effective identifiers (Yang et al., 2023; Sun et al., 2023). A key limitation of these methods is their reliance on task-specific training over designated datasets. With the advent of LLMs, researchers have begun investigating their use as generative models for document retrieval. Some studies propose directly employing LLMs to generate pseudo-documents (Gao et al., 2023a; Mackie et al., 2023; Wang et al., 2023), which are treated as queries in dense retrieval frameworks to locate the final relevant documents. This approach bypasses the need for additional training, instead relying on the generalization capabilities of LLMs. However, LLMs often exhibit content bias, leading to pseudo-documents that poorly align with the target corpus and hinder retrieval performance. To address these challenges, our method focuses on improving the alignment of LLM-generated pseudo-documents with the target document collection. This approach aims to facilitate more accurate and effective downstream document retrieval, improving the overall performance of information retrieval systems.

3 Methodology

In this section, we first introduce the task definition of GAR. Then, we explain how our proposed CA-GAR method is designed.

3.1 Preliminaries

The GAR task consists of two primary steps. First, a large language model (LLM) is prompted with a specific instruction INSTRUCT to generate a text x based on a given query q:

$$x = \text{LLM}(\text{INSTRUCT}, q). \tag{1}$$

Second, the generated text x is utilized as input for a similarity function $\texttt{Sim}$, which measures the similarity between x and each document d. The most similar document d^* from the target corpus D is then identified as:

$$d^* = \arg\max_{d \in D} \texttt{Sim}(x, d). \tag{2}$$

A key limitation of this approach lies in the potential misalignment between the generated text x and the distribution of the target corpus. This misalignment is often caused by the content bias and stochasticity inherent in large language models, which can result in x being poorly aligned with

the characteristics of the target corpus. As a result, the retrieval performance may be negatively affected. Therefore, a critical challenge is to ensure that the generated text x is more closely aligned with the underlying distribution of the target corpus to enhance retrieval effectiveness.

3.2 Context-Aware GAR

Optimization Objective To fully leverage the generative capabilities of LLMs and ensure that the generated text x based on a given query q aligns more closely with the target document, thereby improving its effectiveness in retrieval tasks, we introduce the following optimization objective:

$$\max \mathcal{P}(x) = \sum_{i=1}^{n} \log p_{LM}(x_i) + \beta \cdot \texttt{Sim}(x, d^*), \tag{3}$$

where $\sum_{i=1}^{n} \log p_{LM}(x_i)$ represents the likelihood of the token x_i generated by the LLM, expressed as the sum of the logarithmic probabilities of each individual token, β serves as a tuning factor, controlling the balance between different components of the objective function and $\texttt{Sim}(x, d^*)$ quantifies the degree of similarity or alignment between x and d^*, providing a measure of their relationship.

By integrating these components, the optimization objective encourages the generation of text that is not only coherent and likely under the LLM but also contextually aligned with the target document, thereby improving its effectiveness in downstream retrieval tasks.

Distribution Alignment Strategy However, the document with the highest similarity score is not necessarily the actual ground truth. Therefore, we take multiple candidate documents into consideration. In document retrieval, after measuring the similarity between a query q and a document d using a similarity function $\texttt{Sim}$, we typically obtain the top-k ranked documents, denoted as D_k. This set consists of the k documents with the highest similarity scores relative to q. We define F_k as follows:

$$F_k(x, D) = \frac{1}{k} \sum_{d' \in D_k}^{k} \texttt{Sim}(x, d'), \tag{4}$$

where F_k is no longer a document set but a real-valued function. It represents the average similarity score between a generated text x and the top-k most relevant documents in D.

To more effectively align F_k with information at the x-level and ensure its consistency with the target document collection, we adopt the BM25 method, which demonstrates exceptional generalization ability in zero-shot scenarios. As a lexicon-based retrieval method, BM25 exhibits strong analytical interpretability in similarity computation, with its scoring mechanism explicitly decomposable into the product of inverse document frequency (IDF) and term frequency (TF). Furthermore, BM25 offers significant advantages in interpretability, enabling more precise capture of the characteristics of the target document collection, thereby enhancing retrieval effectiveness and reliability. Therefore, we define the $\texttt{Sim}$ as BM25, allowing Equation 3 to be reformulated as follows:

$$\max \mathcal{P}(x) = \sum_{i=1}^{n} \log p_{LM}(x_i) + \beta \cdot \frac{1}{k} \sum_{d' \in D_k}^{k} \text{IDF}(x) \cdot \text{TF}_{d'}(x), \tag{5}$$

where IDF represents the Inverse Document Frequency, which measures the importance of a term by evaluating how unique or rare it is across the document collection. TF represents the Term Frequency, which quantifies how often a term appears in a specific document.

Referring to the autoregressive decoding method of LLMs, we approximate Equation 5 using Equation 6 as follows:

$$\max \mathcal{P}(x_i|x_{<i}) = \log p_{LM}(x_i|x_{<i}) + \beta \cdot \frac{1}{k} \sum_{d' \in D_{k,i}}^{k} \sum_{j=1}^{i} \text{IDF}(x_j) \cdot \text{TF}_{d'}(x_j), \tag{6}$$

where $D_{k,i}$ represents the top-k documents in D that are most relevant to $x_{0,1,\ldots,i}$. This dynamic selection mechanism allows for an adaptive adjustment of the generation process, ensuring that the most pertinent information is incorporated at each step.

Document Retrieval Based on the generated content x' and the original query q, we construct a new query q' by concatenation:

$$q' = q + x'. \tag{7}$$

We then employ either sparse or dense retrieval methods to retrieve relevant documents.

For sparse retrieval, we utilize the BM25 algorithm, which ranks documents D based on their relevance to the query q' by incorporating TF and IDF. Alternatively, for dense retrieval, we encode both the query q' and document d into vector representations $\mathbf{v_{q'}}$ and $\mathbf{v_d}$ using neural encoders. The similarity score between a query and a document is then computed as the inner product of their respective vectors:

$$\langle E_q(q'), E_d(d) \rangle = \langle \mathbf{v_{q'}}, \mathbf{v_d} \rangle. \quad (8)$$

Regardless of whether sparse or dense retrieval is used, the final retrieval results consist of the top-k documents with the highest similarity scores to the query q'.

4 Experiments

In this section, we provide a detailed explanation of the implementation of CA-GAR and present the experimental results on datasets. Furthermore, we demonstrate the performance improvements achieved by our approach over both BM25 and dense retrieval models.

4.1 Experimental Setup

Datasets and metrics In our main experiments, following previous works (Gao et al., 2023a; Feng et al., 2024), we selected seven low-resource datasets from the BEIR (Thakur et al., 2021) benchmark, covering a diverse range of domains, including biomedical, finance, and scientific research. Additionally, these datasets span various retrieval tasks, such as argument retrieval, citation prediction, and fact-checking. Specifically, the selected datasets include Arguana (Wachsmuth et al., 2018), Scifact (Wadden et al., 2020), NFCorpus (Boteva et al., 2016), Scidocs (Cohan et al., 2020), FiQA (Maia et al., 2018), TrecCovid (Voorhees et al., 2020), and Touché (Bondarenko et al., 2020). We use nDCG@10 as the evaluation metric.

To evaluate the effectiveness of our approach in a multilingual setting, we conduct experiments on non-English datasets by selecting four low-resource languages from the Mr.TyDi (Zhang et al., 2021) benchmark: Bengali, Swahili, Telugu, and Thai. For performance assessment, we employ nDCG@10 as the evaluation metric.

For different datasets, we employed tailored prompt instructions to generate more appropriate content. In subsequent comparative experiments, we consistently used the same prompt to ensure fairness and reliability. Detailed instructions can be found in the appendix A.

Implementation details In this study, we utilize LLaMA3-8B-Instruct (Dubey et al., 2024) as the large language model for content generation. For retrieval, we employ BM25 as the sparse retrieval model, implemented using the BM25S (Lù, 2024) library, which is both highly efficient and simple. For English datasets, we apply the corresponding English stemmer to enhance retrieval performance. However, for multilingual datasets, we adopt a unified approach by not applying stemming to ensure consistency across different languages. The dense retrieval models include Contriever (Izacard et al., 2021), Contriever-ft (fine-tuned on MS MARCO), and BGE (Xiao et al., 2024). For non-English datasets, we similarly adopt BM25 as the sparse retrieval model without applying a stemmer. The dense retrieval models for these datasets include mContriever and mContriever-ft (fine-tuned on MS MARCO). In our experiment, we configure the LLaMA3-8B-Instruct model with a temperature of 1.0, a top-p of 1.0, and a top-k of 50. The parameter k in F_k is set to 10 and the β parameter is selected from {0.25,0.5,0.75,1.0} based on the best performance, and it is set to 0.75. The BM25 parameters k_1 and b are configured as 1.5 and 0.75, respectively. All the experiments are conducted with the single A800 GPU with 80GB VRAM.

Baselines First, we consider the traditional term frequency-based sparse retrieval method, BM25, which has been widely recognized for its effectiveness in zero-shot scenarios, learned sparse retrieval method DeepCT (Dai and Callan, 2020) and docT5query (Nogueira et al., 2019). Next, we examine Contriever for English and mContriever for non-English languages, along with their fine-tuned versions on the MS MARCO dataset, referred to as Contriever-ft and mContriever-ft, respectively. Additionally, we incorporate several dense retrieval models, including DPR (Karpukhin et al., 2020), ANCE (Xiong et al., 2021), latent interaction-based ColBERT (Khattab and Zaharia, 2020) and ColBERTv2 (Santhanam et al., 2022). Furthermore, we include BGE, which has demonstrated strong performance on English datasets.

Furthermore, we compare our approach with HyDE. To ensure a fair comparison in terms of LLM selection, we use LLaMA3-8B-Instruct to generate hypothetical documents. Following the

Model	Avg	Arguana	Scifact	NFCorpus	Scidocs	FiQA	Trec-Covid	Touché
Baselines (*Prior Work*)								
DeepCT	30.2	30.9	63.0	30.1	12.4	19.1	40.6	15.6
docT5query	40.9	34.9	67.5	32.8	16.2	29.1	71.3	34.7
DPR	19.1	17.5	31.8	18.9	7.7	11.2	33.2	13.1
ANCE	35.3	41.5	50.7	23.7	12.2	29.5	65.4	24.0
ColBERT	36.4	23.3	67.1	30.5	14.5	31.7	67.7	20.2
ColBERTv2	42.9	46.3	69.3	33.8	15.4	35.6	73.8	26.3
BM25	40.1	39.7	66.5	32.5	15.8	23.6	65.6	36.7
Contriever	32.8	37.9	64.9	31.7	13.7	24.5	36.3	20.8
Contriever-ft	40.3	44.6	67.7	32.8	16.5	32.9	59.6	27.8
BGE	48.6	63.5	74.6	**38.1**	22.6	44.3	72.3	24.8
HyDE$_{llama3\text{-}8b}$								
w/ Contriever	34.6	34.1	63.9	31.4	13.8	26.5	46.3	26.4
w/ BGE	48.7	66.0	74.4	36.9	20.1	41.0	**75.6**	26.8
CA-GAR$_{llama3\text{-}8b}$ (*Our Work*)								
w/ BM25	$42.2^{\uparrow 2.1}$	40.5	67.7	33.1	15.4	23.7	66.4	**48.3**
w/ Contriever	$34.9^{\uparrow 2.1}$	38.1	64.5	30.8	14.1	24.9	45.8	26.5
w/ Contriever-ft	$41.8^{\uparrow 1.5}$	45.2	67.8	33.2	17.2	33.2	60.3	35.9
w/ BGE	$\mathbf{50.0}^{\uparrow 1.4}$	**69.3**	**75.5**	35.4	**22.8**	**44.4**	72.1	30.2

Table 1: **Low-resource retrieval performance on a selection BEIR tasks** (measured by nDCG@10).

Model	Avg	bn	sw	te	th
Baselines (*Prior Work*)					
BM25	30.3	19.1	48.8	14.9	38.4
mContriever	19.5	21.3	23.9	10.7	21.9
HyDE$_{llama3\text{-}8b}$	24.1	32.4	24.8	14.2	25.1
mContriever-ft	48.2	46.8	57.6	44.2	44.0
CA-GAR$_{llama3\text{-}8b}$ (*Our Work*)					
w/ BM25	$34.9^{\uparrow 4.6}$	25.3	50.8	20.7	42.9
w/ mContriever	$24.7^{\uparrow 5.2}$	32.9	25.4	14.8	25.6
w/ mContriever-ft	$\mathbf{59.2}^{\uparrow 11.0}$	**55.5**	**62.1**	**64.9**	**54.2**

Table 2: **Multi-lingual retrieval performance on a selection Mr.TyDi languages** (measured by nDCG@10).

HyDE methodology, we employ Contriever and mContriever as the retrieval model, enabling a comparison between the approaches.

4.2 Low-Resource Retrieval

In Table 1, we present the performance of CA-GAR across seven tasks selected from the BEIR dataset. The results indicate that, on average, CA-GAR achieves a noticeable improvement in nDCG@10 compared to each baseline. In particular, the improvement over BM25 is especially pronounced. This can likely be attributed to our approach of leveraging BM25 to retrieve relevant documents during the decoding process of the large language model (LLM). Furthermore, compared to HyDE, CA-GAR achieves a certain degree of improvement when using Contriever as the retriever.

Our method achieves notable improvements on the Touché dataset, likely due to BM25's superior performance compared to other retrievers, including Contriever, Contriever-ft, and BGE. Since BM25 significantly influences LLM-generated responses, its strong baseline on Touché contributes to the observed performance gains. In contrast, improvements on FiQA and Scidocs are modest, primarily because BM25 performs poorly on these datasets, limiting its impact on LLM's autoregressive decoding. Additionally, these datasets involve complex domains (e.g., finance and scientific citation prediction), where LLM-generated responses tend to be of lower quality. A similar trend is observed for NFCorpus, which focuses on biomedical information retrieval. In such cases, more advanced prompting strategies or LLM architectures may be

Model	Avg	Arguana	Scifact	NFCorpus	Scidocs	FiQA	Trec-Covid	Touché
BM25								
CA-GAR$_{llama3\text{-}8b}$	42.2	40.5	67.7	33.1	15.4	23.7	66.4	48.3
w/o CA	41.6$^{\downarrow 0.6}$	39.8	66.9	32.1	14.8	23.5	66.9	47.3
Contriever								
CA-GAR$_{llama3\text{-}8b}$	34.9	38.1	64.5	30.8	14.1	24.9	45.8	26.5
w/o CA	34.2$^{\downarrow 0.7}$	37.6	64.3	29.7	13.1	23.9	45.5	25.3
Contriever-ft								
CA-GAR$_{llama3\text{-}8b}$	41.8	45.2	67.8	33.2	17.2	33.2	60.3	35.9
w/o CA	40.9$^{\downarrow 0.9}$	44.5	64.8	32.9	16.3	31.9	59.8	35.8
BGE								
CA-GAR$_{llama3\text{-}8b}$	50.0	69.3	75.5	35.4	22.8	44.4	72.1	30.2
w/o CA	49.0$^{\downarrow 1.0}$	68.7	73.8	35.3	21.7	43.9	70.5	29.3

Table 3: **Ablation study on a selection BEIR tasks** (measured by nDCG@10).

Model	Avg	bn	sw	te	th
BM25					
CA-GAR$_{llama3\text{-}8b}$	34.9	25.3	50.8	20.7	42.9
w/o CA	34.2$^{\downarrow 0.7}$	24.9	50.1	19.8	42.1
mContriever					
CA-GAR$_{llama3\text{-}8b}$	24.7	32.9	25.4	14.8	25.6
w/o CA	23.9$^{\downarrow 0.8}$	32.2	24.5	14.0	24.7
mContriever-ft					
CA-GAR$_{llama3\text{-}8b}$	59.2	55.5	62.1	64.9	54.2
w/o CA	58.2$^{\downarrow 1.0}$	54.9	61.2	64.1	52.7

Table 4: **Ablation study on a selection Mr.TyDi languages** (measured by nDCG@10).

required for further improvements.

4.3 Multi-Lingual Retrieval

In Table 2, we present the performance results on four low-resource languages selected from the Mr.TyDi dataset. Our findings indicate that CA-GAR consistently outperforms the baseline across four languages. The performance improvement can be attributed not only to the robust multilingual retrieval capabilities of BM25, which enhance the final content generated by the LLM, but also to the inherently strong multilingual proficiency of LLaMA 3, which further contributes to the observed performance gains.

Additionally, for the stronger retrieval model, mContriever-ft, the generated content tends to be richer in information. As a result, more powerful models are likely to achieve greater performance improvements, as they are less affected by biases introduced by extraneous information. Compared to HyDE, CA-GAR with Contriever also demonstrates a noticeable improvement, suggesting that BM25 plays an effective role in influencing the LLM's autoregressive decoding process.

4.4 Ablation Study

To further validate the effectiveness of our CA-GAR approach, we conducted an ablation study on selected datasets from BEIR and Mr.TyDi. The detailed experimental results are presented in Table 3 and 4. Specifically, we compared CA-GAR with a variant that does not employ the context-aware mechanism, denoted as w/o CA. In this baseline, the LLM generates content in a straightforward manner, without incorporating context-aware strategies to guide the autoregressive decoding process. This comparison allows us to assess the impact of context-aware strategies on the model's overall performance and the quality of the generated outputs.

Our findings indicate that, compared to CA-GAR, w/o CA exhibits a noticeable decline in overall performance. This performance degradation is particularly pronounced for relatively stronger models such as Contriever-ft, mContriever-ft and BGE. The results suggest that when the generated content exhibits certain biases or misalignments, the adverse effects are more pronounced in these high-performing models. This observation further highlights the critical role of CA-GAR in guiding and refining the generated content.

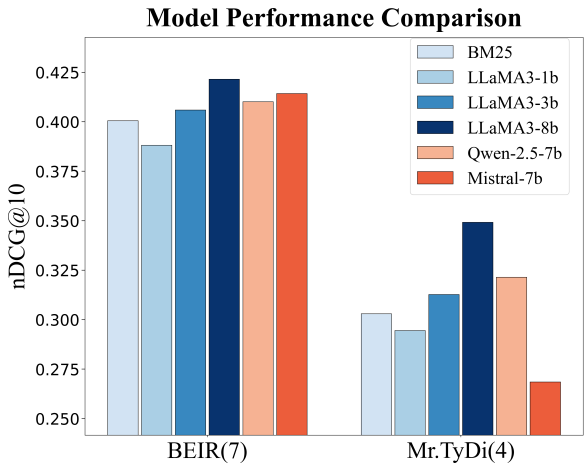

Figure 3: A figure presenting the retrieval performance of different models using BM25 across seven tasks from the BEIR(7) dataset and four languages from the Mr.TyDi(4) dataset.

5 Analysis

5.1 LLM Base Model Comparsion

To investigate the impact of model size, we compare three LLaMA3-Instruct variants with 1B, 3B, and 8B parameters. Figure 3 shows their performance when integrated with BM25.

Our results demonstrate that larger models generally achieve superior performance due to their enhanced generative capabilities. In contrast, smaller models, the 1B variant, often introduce substantial deviations in generated content, adversely affecting retrieval effectiveness. The 3B model performs on par with the baseline, exhibiting a slight improvement. These findings highlight the critical role of generative model quality in retrieval performance, as lower-quality models may introduce biases or hallucinated content that degrade results.

Additionally, we selected other LLMs of comparable scale, Qwen-2.5-7B (Yang et al., 2024) and Mistral-7B (Jiang et al., 2023), and conducted experiments on the dataset. It can be observed that among the seven datasets selected from BEIR, LLaMA3-8B-Instruct achieves the highest average performance, while LLaMA3-8B-Instruct and Mistral-7B perform similarly, with LLaMA3-8B-Instruct being slightly better. In contrast, Qwen-2.5-7B performs relatively poorly.

Furthermore, in multilingual settings, LLaMA3-8B-Instruct's superior capabilities enhance retrieval performance, while Mistral-7B's weaker multilingual abilities lead to greater deviations, resulting in lower performance than BM25.

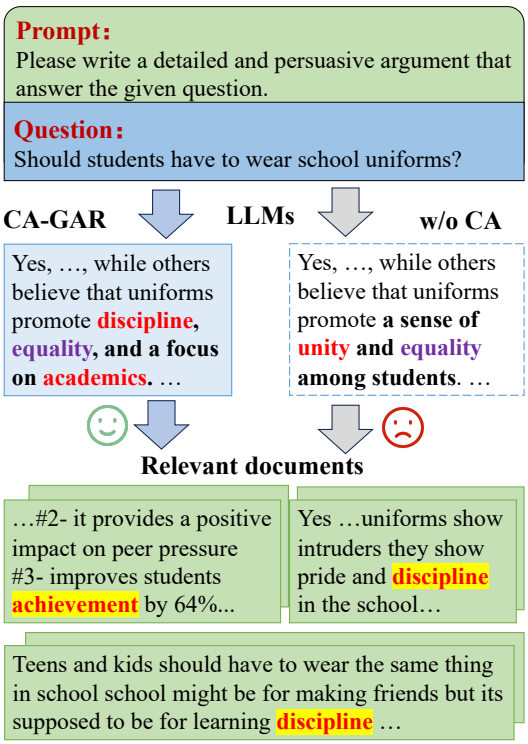

Figure 4: Case Study: An example query from Touché – A comparison of content generated by CA-GAR and w/o CA.

5.2 Case Study

To further demonstrate the effectiveness of our approach, we selected a specific question from the Touché dataset as a case study. As illustrated in Figure 4, when responding to the same prompt for the question *"Should students have to wear school uniforms?"*, both our method, CA-GAR, and w/o CA generate relevant responses that address the question and included the keyword *"equal"*. However, there are notable differences in the choice of other words. CA-GAR incorporated *"discipline"* and *"academics"*, whereas w/o CA used *"unity"* .

From a generative standpoint, both outputs are acceptable. However, for subsequent retrieval processes, the content generated by CA-GAR proves to be superior. This is because it includes semantically related terms such as *"academics"* and *"achievement"*, as well as the identical term *"discipline"*, which are more aligned with the relevant documents in the corpus. This indicates that our context-aware approach successfully influences the autoregressive decoding process of large language models, resulting in generated content that is more aligned with the target retrieval corpus.

β	Arguana	Scifact	NFCorpus	Scidocs	FiQA	Trec-Covid	Touché	Avg
0.25	39.88	67.43	33.10	15.39	23.72	66.54	47.69	41.96
0.5	40.02	67.45	33.14	15.38	23.62	66.28	48.29	42.03
0.75	40.50	67.70	33.14	15.38	23.69	66.43	48.29	**42.16**
1.0	39.70	67.64	33.14	15.38	23.62	66.43	48.19	42.01

Table 5: Performance results for different values of β on a selection BEIR tasks.

5.3 Hyperparameter Analysis

In our study, we address the integration of log probabilities and BM25 scores by introducing a scalar weighting factor, denoted as β. This factor is pivotal in modulating the influence of BM25-derived guidance relative to the model's intrinsic generation probabilities. To identify the optimal value of β, we conducted a series of experiments across multiple datasets, systematically varying β to assess its impact on model performance.

The results are summarized in Table 5. From the results, it is evident that the value of $\beta = 0.75$ yields the highest average performance across all datasets. This suggests that a balanced approach, where the influence of BM25 scores is moderately weighted, enhances the model's effectiveness in generating relevant outputs.

6 Conclusion

This paper introduces CA-GAR, a novel generative augmentation retrieval method that optimizes LLMs generation using a distribution alignment strategy. By leveraging a lexicon-based approach with BM25, CA-GAR ensures better alignment between generated content and target documents. Experiments on low-resource and multilingual datasets validate its effectiveness, with advanced LLMs further enhancing performance. A case study illustrates CA-GAR's impact on content generation, highlighting its ability to maximize LLMs' generative potential.

Limitations

Our approach relies on LLMs to generate content while incorporating context-awareness to influence the autoregressive decoding process during generation. However, this method may not be well-suited for scenarios requiring low latency. Nevertheless, advancements in hardware and the development of optimization algorithms for model inference are expected to significantly reduce the computational cost and latency of content generation. Moreover, future research could explore alternative approaches to influencing the autoregressive decoding process of LLMs, beyond the use of BM25. This opens the possibility for more generalized and adaptable methods to enhance generation quality and relevance.

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A Instructions

Prompts for Arguana

Please write a counter argument for the passage.
Passage: {PASSAGE}
Counter Argument:

Prompts for Scifact

Please write a scientific paper passage to support or refute the claim.
Claim: {CLAIM}
Passage:

Prompts for NFCorpus

Please write a medically accurate passage to answer the question.
Question: {QUESTION}
Passage:

Prompts for Scidocs

Please write a scientific paper abstract that is cited by the given scientific paper title.
Paper title: {PAPER TITLE}
Paper abstract:

Prompts for FiQA

Please write a financial article passage to answer the question.
Question: {QUESTION}
Passage:

Prompts for Trec-Covid

Please write a passage that answer the question on COVID-19.
Question: {QUESTION}
Passage:

Prompts for Touché

Please write a detailed and persuasive argument that answer the given question.
Question: {QUESTION}
Argument:

Prompts for Mr.TyDi

Please write a passage in {LANGUAGE} to answer the question in detail.
Question: {QUESTION}
Passage:

B Analysis for efficiency

Our method introduces a slight increase in latency due to the relevance-guided generation process. In our experiments on the Scifact dataset, which consists of a corpus of 5k documents and 300 queries, we utilized a single NVIDIA A800 GPU and employed the LLaMA-3-8B model for document generation. The latency per query for each method is summarized in Table 6.

Method	Latency
HyDE	2.059s
CA-GAR	2.933s

Table 6: Latency per query on Scifact

The observed latency overhead primarily arises from the incorporation of document-level signals during the generation process. When the corpus size increases, applying CA-GAR over the entire corpus can lead to substantial latency. However, in practical applications, it is feasible to first retrieve a small subset of documents relevant to each query, which can effectively guide the generation process.

To evaluate this strategy, we conducted experiments on the large-scale Touché dataset, which contains 382k documents. The results, summarized in Table 7:

Method	Latency	nDCG@10
HyDE	2.498s	26.8
CA-GAR		
w/ 1k subset	2.635s	29.5
w/ 3k subset	3.134s	29.8
w/ 5k subset	3.621s	29.9
w/ full corpus	79.923s	30.2

Table 7: Latency and nDCG@10 on Touché

The results indicate that utilizing a pre-filtered subset of documents (e.g., the top 1k or 3k documents ranked by BM25) significantly reduces latency while maintaining competitive performance metrics. This approach enhances the practicality and scalability of CA-GAR in real-world retrieval systems. In conclusion, while there is a noted efficiency overhead associated with CA-GAR, this does not detract from its core contributions.